



BLINKPAY

An Offline NFC-Based Payment and Communication Application

Presented by

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Problem Statement

The Problem

- Most digital payment workflows depend on internet connectivity.
- Payment and communication are usually handled by separate applications.
- Poor connectivity can prevent or delay transactions.

Our Idea

BlinkPay combines payment + communication in one Android application, with offline payment through NFC.

Key Features

Digital Wallet

- Send / receive money
- Wallet balance
- Transaction history

Offline NFC Payment

- Device-to-device payment
- Works without active internet connectivity

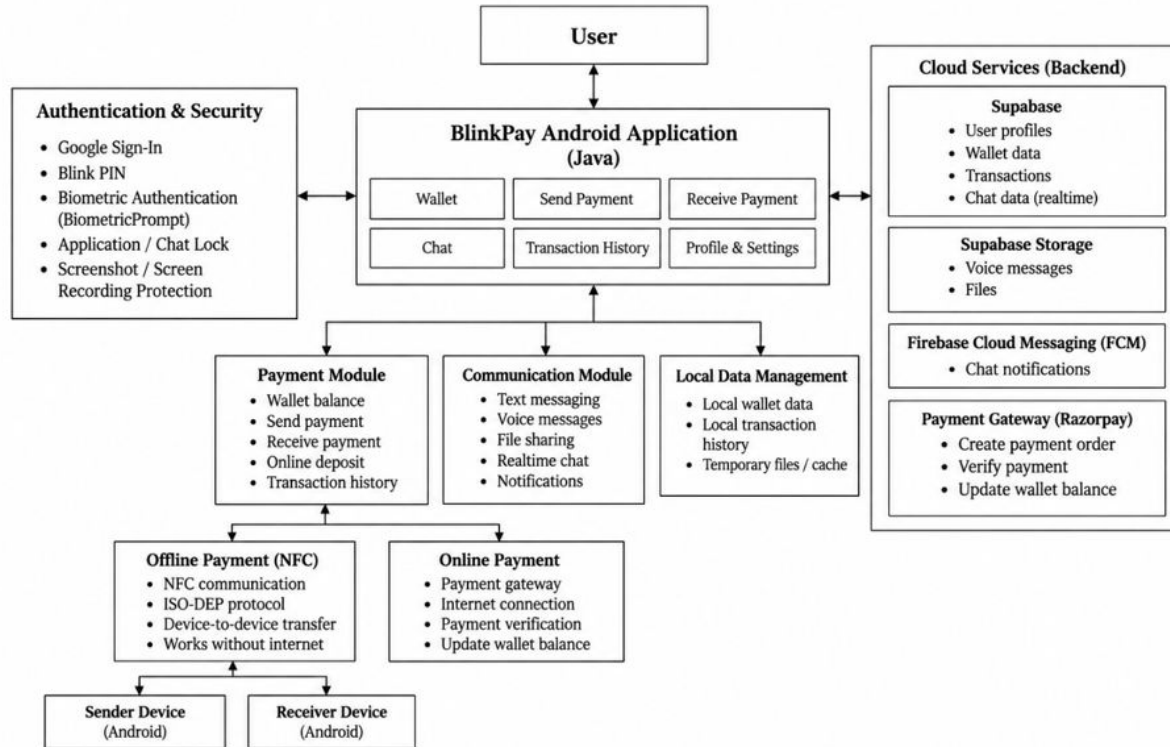
Security

- Google Sign-In
- 4-digit Blink PIN
- Biometric authentication

Communication

- Real-time text chat
- Voice messages
- File sharing
- Push notifications

System Architecture



Technology Stack

Area	Technology
Language	Java
Platform	Android
NFC	NfcAdapter, Tag, IsoDep
Authentication	Credential Manager, Biometric Prompt
Backend	Supabase
Notifications	Firebase Cloud Messaging
Online Payment	Razorpay SDK
Networking	OkHttp

Implementation & Results

Implementation

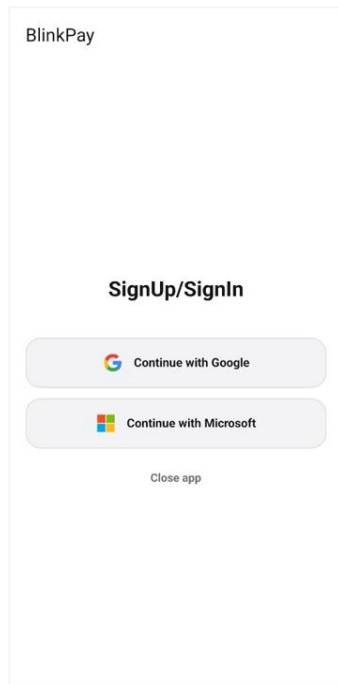
- Modular Java + Android architecture
- Payment amount validation
- SHA-256 PIN hashing
- NFC sender/receiver workflow
- Local wallet handling
- Cloud synchronization
- Chat and notifications

Results

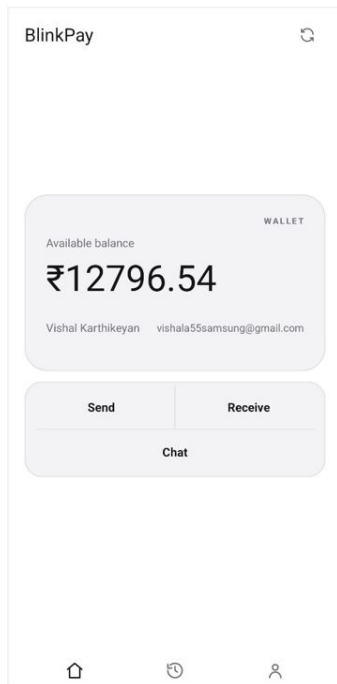
19 / 19 test cases passed

Average observed response time: 0.412 seconds

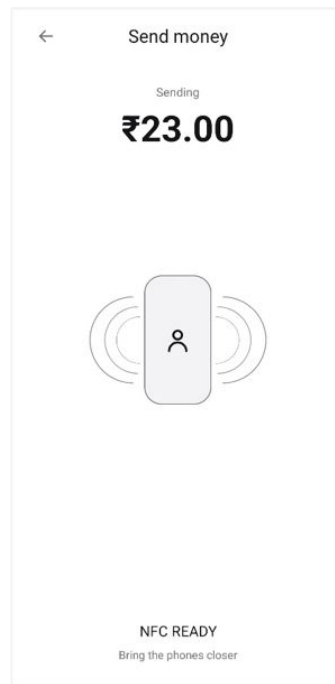
Screenshots



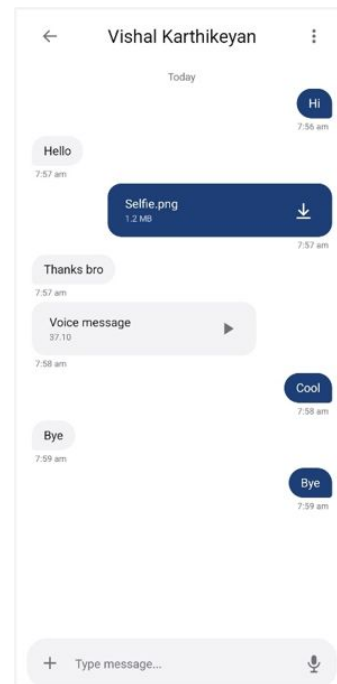
Auth Screen



Wallet Screen



NFC Screen



Chat Screen

Conclusion & Future Scope

Conclusion

BlinkPay successfully demonstrates a unified Android application combining:

Offline NFC Payment + Online Payment + Wallet + Security + Communication

Future Scope

- Improve NFC reliability across more devices
- Advanced transaction management
- Stronger authentication options
- Enhanced communication features
- Large-scale and multi-user testing
- Production-oriented backend and deployment